

General self-dual Yang-Mills solutions

Norman H. Christ and Erick J. Weinberg

Department of Physics, Columbia University, New York, New York

Nancy K. Stanton

Department of Mathematics, Columbia University, New York, New York

(Received 5 May 1978)

The recent work of Atiyah, Hitchin, Drinfeld, and Manin is used to discuss self-dual Yang-Mills solutions for the compact gauge groups $O(n)$, $SU(n)$, and $Sp(n)$. It is shown that the resulting solutions contain the correct number of parameters for all values of the topological charge. Although explicit construction of a general self-dual field requires the solution of a finite-dimensional, nonlinear matrix equation, we show that for widely separated instantons this equation can be solved perturbatively, providing a systematic expansion about the dilute-gas limit and a physical interpretation of the independent parameters in this limit. Further, closed-form expressions can be obtained for the general $SU(2)$ solutions with topological charge 2 or 3. Finally, explicit isospin-1/2 and isospin-1 propagators are derived for a massless scalar field in the presence of the general self-dual $SU(2)$ solution.

I. INTRODUCTION

A significant advance in the study of self-dual Yang-Mills fields has recently been made by Atiyah, Hitchin, Drinfeld, and Manin.^{1,2} These authors apply a construction due to Horrocks and Barth³ to the complex formulation of self-dual Yang-Mills fields invented by Ward.^{4,5} They reduce the nonlinear partial differential equations of self-duality to a nonlinear equation for a finite-dimensional matrix. The resulting $SU(2)$ solutions contain the correct number of independent parameters and can be generalized in a straightforward way to $Sp(n)$, $SU(n)$, and $O(n)$. Finally Atiyah, Hitchin, Drinfeld, and Manin argue that all self-dual fields can be obtained by this procedure.

In this paper we present the construction of the above authors as a remarkably simple *Ansatz* and show directly that the resulting gauge field is self-dual. This is done for $SU(2)$ in Sec. II and explicitly generalized to $O(n)$, $SU(n)$, and $Sp(n)$ in Sec. III. Next, in Sec. IV, we introduce a canonical form for the matrices appearing in the *Ansatz* and determine the number of independent parameters present in the $O(n)$, $SU(n)$, and $Sp(n)$ solutions for arbitrary topological charge. The results agree with those of the previous analysis⁶ based on the Atiyah-Singer index theorem. In Sec. V the above construction is applied to obtain both the familiar $SU(2)$ 't Hooft k -instanton solution and an explicit, systematic expansion of the general $SU(2)$ solution with topological charge k in the limit of large instanton separations. These results are generalized to $O(n)$, $SU(n)$, and $Sp(n)$ in Sec. VI. Finally, in the concluding section we discuss the general self-dual $SU(2)$ solutions for $k=2$ and 3 and present the

isospin- $\frac{1}{2}$ and isospin-1 propagators for a scalar field in the presence of the general $SU(2)$ solutions.

II. $SU(2)$ CONSTRUCTION

We wish to construct a self-dual Yang-Mills field A_μ with topological charge

$$k = -\frac{1}{16\pi^2} \int d^4x \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) \quad (2.1)$$

and containing $8k - 3$ independent parameters. Here the Yang-Mills field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \quad (2.2)$$

and the gauge field A_μ is viewed as a 2×2 , anti-Hermitian, traceless matrix.

Such solutions can be constructed from a $(k+1) \times k$ matrix $M(x)$ made up of quaternions. Thus an element M_{ab} of the matrix M can be written in terms of four real numbers M_{ab}^μ , $0 \leq \mu \leq 3$, and the unit quaternions i , j , and $\hat{k}$:

$$M_{ab} = M_{ab}^0 + M_{ab}^1 i + M_{ab}^2 \hat{j} + M_{ab}^3 \hat{k}. \quad (2.3)$$

The quantities $\hat{i}$, $\hat{j}$, and $\hat{k}$ anticommute and obey

$$\hat{i}^2 = \hat{j}^2 = \hat{k}^2 = -1, \quad \hat{i} \times \hat{j} = \hat{k}.$$

Occasionally it will be helpful to treat each quaternionic element M_{ab} as a 2×2 matrix

$$M_{ab} = \sum_{\mu=0}^3 \sigma_\mu M_{ab}^\mu, \quad (2.5)$$

where σ_0 is the 2×2 identity matrix and $\sigma_j = -i\tau^j$, $1 \leq j \leq 3$, τ^j being the j th Pauli matrix. The matrix M is chosen to be linear in x ,

$$M = B - Cx, \quad (2.6)$$

where B and C are constant $(k+1) \times k$ quaternionic matrices of rank k (Ref. 7) and x is the quaternion

$$x = x_0 + x_1 \hat{i} + x_2 \hat{j} + x_3 \hat{k} \quad (2.7)$$

formed from the four Euclidean position coordinates x_μ . Finally $M(x)$ is assumed to satisfy the nonlinear requirement

$$M^\dagger(x)M(x) = R, \quad (2.8)$$

a real, invertible $k \times k$ matrix. Here $M^\dagger$ is the transpose of the quaternionic conjugate of M :

$$(M^\dagger)_{ij}^a = M_{ji}^a, \quad (M^\dagger)_{ij}^a = -M_{ji}^a, \quad 1 \leq a \leq 3.$$

In order to construct a self-dual gauge field $A_\mu(x)$ from $M(x)$ it is necessary to find a $(k+1)$ -dimensional column vector $N(x)$ obeying

$$N^\dagger(x)M(x) = 0, \quad (2.9a)$$

$$N^\dagger(x)N(x) = 1. \quad (2.9b)$$

The linear equation (2.9a) can be viewed as k quaternionic conditions on the $k+1$ elements of N . Thus a solution $N(x)$ to Eq. (2.9a) can always be found and the requirement (2.9b) simply fixes its normalization. The gauge field $A_\mu(x)$ is now defined by

$$A_\mu(x) = N^\dagger(x) \partial_\mu N(x), \quad (2.10)$$

where $A_\mu(x)$ can be identified as a 2×2 matrix if the quaternion on the right-hand side of Eq. (2.10) is considered to be a 2×2 matrix as is done in Eq. (2.5). Differentiating the normalization condition (2.9b), one easily sees that $A_\mu(x)$ is anti-Hermitian:

$$A_\mu(x)^\dagger = \partial_\mu N^\dagger(x)N(x) = -N^\dagger(x) \partial_\mu N(x). \quad (2.11)$$

Furthermore, any anti-Hermitian 2×2 matrix which has the quaternionic form (2.5) must necessarily be traceless.

Thus Eq. (2.10) defines an $SU(2)$ gauge potential. We will now show that the resulting field strength $F_{\mu\nu}$ is self-dual. Substituting into Eq. (2.2) one obtains

$$F_{\mu\nu} = (\partial_\mu N^\dagger)(\partial_\nu N) + (N^\dagger \partial_\mu N)(N^\dagger \partial_\nu N) - [\mu \leftrightarrow \nu] \quad (2.12)$$

or, using the right-hand part of Eq. (2.11) twice,

$$F_{\mu\nu} = \partial_\mu N^\dagger [I - NN^\dagger] \partial_\nu N - [\mu \leftrightarrow \nu]. \quad (2.13)$$

The quantity in curly brackets in Eq. (2.13) is simply the projection operator onto the k -dimensional quaternionic subspace orthogonal to N . Using Eqs. (2.9a) and (2.8) this can be written as the $(k+1) \times (k+1)$ matrix

$$I - NN^\dagger = M(x)R^{-1}(x)M^\dagger(x), \quad (2.14)$$

where R^{-1} is the $k \times k$ matrix inverse of the real matrix R in Eq. (2.8). [It is easy to see that the

matrix (2.14) is equal to its square.] Thus, using

$$(\partial_\mu N^\dagger)M = -N^\dagger \partial_\mu M \quad (2.15)$$

[a consequence of Eq. (2.9a)], $F_{\mu\nu}$ can be written

$$\begin{aligned} F_{\mu\nu} &= N^\dagger (\partial_\mu M) R^{-1} (\partial_\nu M^\dagger) N - [\mu \leftrightarrow \nu] \\ &= N^\dagger C (\sigma_\mu R^{-1} \sigma_\nu^\dagger - \sigma_\nu R^{-1} \sigma_\mu^\dagger) C^\dagger N, \end{aligned} \quad (2.16)$$

in the notation of Eqs. (2.5) and (2.6). Since the elements of the matrix R^{-1} are real, they commute with the "quaternions" σ_μ and $\sigma_\nu^\dagger$, and $F_{\mu\nu}$ explicitly contains the self-dual combination

$$\sigma_\mu \sigma_\nu^\dagger - \sigma_\nu \sigma_\mu^\dagger = 2i \tau^a \eta_{a\mu\nu}, \quad (2.17)$$

where $\eta_{a\mu\nu}$, $1 \leq a \leq 3$ are similar to the three self-dual matrices defined by 't Hooft.⁸

As will be shown in Secs. IV and V, the solution (2.10) possesses topological charge k and contains precisely $8k - 3$ independent parameters. Thus this surprisingly simple procedure has produced a general self-dual solution. Of course to obtain explicit solutions one must find matrices M obeying the quadratic condition (2.9a). This can be done easily for $k=1, 2$, and 3 , and yields genuinely new $k=3$ solutions. In Sec. V we show how the familiar $SU(2)$ solutions of 't Hooft arise using this procedure and derive a method for obtaining approximate solutions in the limit of small, widely separated instantons for arbitrary k .

III. $O(n)$, $SU(n)$, AND $Sp(n)$ CONSTRUCTIONS

Let us now generalize the *Ansatz* of the previous section to give self-dual $O(n)$ solutions. We will then impose restrictions on that *Ansatz* to reduce our $O(2n)$ solutions to $SU(n)$ solutions and further conditions to specialize $O(4n)$ solutions to $Sp(n)$. The $Sp(1)$ case will then be the simple *Ansatz* described above [recall $Sp(1) = SU(2)$].

$O(n)$. Let $M(x)$ be a $(4k+n) \times k$ matrix of quaternions linear in x ,

$$M(x) = B - Cx, \quad (3.1)$$

where B and C are both $(4k+n) \times k$ constant quaternionic matrices of rank k . We require that the product $M^\dagger(x)qM(x)$ be a real $k \times k$ matrix for any choice of the quaternion q . Thus we can define a $k \times k$ quaternionic matrix R by the equation

$$\sum_{i=1}^{4k+n} [M(x)^\dagger]_{ji} q M(x)_{ii} = 4 \operatorname{Re}(R_{ij} q), \quad (3.2)$$

valid for any quaternion q . The quantity $\operatorname{Re}(R_{ij} q)$ is the real part of the enclosed quaternion $R_{ij} q$, i.e., one-half of the trace of the quaternion $R_{ij} q$ when viewed as a 2×2 matrix. It follows from the definition (3.2) that $R = R^\dagger$; in addition, R is required to be invertible.

Next we introduce n real $(4k+n)$ -dimensional vectors N_i , $1 \leq i \leq n$, with components N_{ji} , $1 \leq j \leq 4k+n$. These vectors are chosen to be orthonormal,

$$\sum_{i=1}^{4k+n} N_{ji} N_{ij} = \delta_{ij}, \quad (3.3a)$$

and to satisfy the $4k$ real conditions

$$\sum_{i=1}^{4k+n} N_{ji} M_{ij} = 0, \quad 1 \leq j \leq k. \quad (3.3b)$$

The resulting components N_{ij} can be viewed as a $(4k+n) \times n$ matrix obeying

$$N^t N = I, \quad (3.4a)$$

$$N^t M = 0, \quad (3.4b)$$

where I represents the $n \times n$ unit matrix. Finally the $O(n)$ vector potential $A_\mu(x)$ is defined by

$$A_\mu(x) = N^t \partial_\mu N. \quad (3.5)$$

The orthogonality condition (3.4a) directly implies that $A_\mu(x)$ is an antisymmetric matrix and therefore properly lies in the adjoint representation of $O(n)$.

Just as in Sec II, we can demonstrate the self-duality of A_μ by studying the field strength $F_{\mu\nu}$ which, as before, can be written

$$F_{\mu\nu} = \partial_\mu N^t \{I - NN^t\} \partial_\nu N - [\mu \leftrightarrow \nu]. \quad (3.6)$$

Again the $(4k+n) \times (4k+n)$ matrix in curly brackets can be viewed as a projection operator onto the $4k$ -dimensional subspace orthogonal to the n columns of N . This subspace and consequently the projection operator $I - NN^t$ can be directly constructed from the k independent quaternionic columns of M provided their normalization, Eq. (3.2), is taken into account. The result is

$$(I - NN^t)_{ij} = \sum_{l=1}^k \text{Re}(M_{il}^* R_{lj}^{-1} M_{il}). \quad (3.7)$$

[Equation (3.2) can be used to show that the right-hand side of Eq. (3.7) is equal to its square. This calculation is most easily done by using the 2×2 matrix form for the quaternions.] In a more compact notation,

$$I - NN^t = \text{Re}(M^* R^{-1} M^t), \quad (3.8)$$

where M^* is the $(4k+n) \times k$ matrix whose elements are the quaternionic conjugates of those in M .

[Note that with noncommuting quaternions $(AB)^t \neq B^t A^t$.] Substituting the projection operator (3.8) into Eq. (3.6) for $F_{\mu\nu}$ and using $(\partial_\mu N^t)M = -N^t(\partial_\mu M)$, one obtains

$$\begin{aligned} F_{\mu\nu} &= N^t \text{Re}(\partial_\mu M^* R^{-1} \partial_\nu M^t) N - [\mu \leftrightarrow \nu] \\ &= N^t \text{Re}(\sigma_\mu^* C^* R^{-1} C^t \sigma_\nu) N - [\mu \leftrightarrow \nu] \\ &= N^t \text{Re}[C^* R^{-1} C^t (\sigma_\mu \sigma_\nu^* - \sigma_\nu \sigma_\mu^*)] N, \end{aligned} \quad (3.9)$$

which from Eq. (2.17) is explicitly self-dual. We have thus obtained a self-dual $O(n)$ solution. It will be shown in Sec. IV that it in fact contains the right number of independent parameters.

SU(n). We can construct self-dual $SU(n)$ solutions by beginning with the $O(2n)$ solutions found above and imposing a constraint. This constraint is designed to guarantee that the effect of the $2n \times 2n$ matrix A_μ is the same as that of an $n \times n$ Hermitian matrix A_μ^C acting on n -dimensional complex vectors. The real and imaginary parts of these complex vectors form the $2n$ real dimensions on which the original matrix A_μ operates.

Thus one begins with a $(4k+2n) \times k$ matrix of quaternions $M(x)$ linear in x and obeying Eq. (3.2). In addition, one requires

$$JM(x) = \hat{i}M(x), \quad (3.10)$$

where J is a $(4k+2n) \times (4k+2n)$ matrix made up of four $(2k+n) \times (2k+n)$ blocks

$$J = \begin{pmatrix} 0 & +I \\ -I & 0 \end{pmatrix}. \quad (3.11)$$

The condition (3.11) requires that the lower $2k+n$ rows of M be simply $\hat{i}$ times the upper $2k+n$ rows,

$$M(x) = \begin{pmatrix} M^C(x) \\ \hat{i}M^C(x) \end{pmatrix}, \quad (3.12)$$

where $M^C(x)$ is a $(2k+n) \times k$ quaternionic matrix.

Using the special form (3.12), the previous equations for the $O(n)$ case can be significantly simplified. In terms of the matrix M^C , the condition (3.2) becomes

$$\sum_{i=1}^{2k+n} (M^C(x)^*)_{ji} (q - \hat{i}q\hat{i}) M_{ii}^C = 4 \text{Re}(R_{ij}q). \quad (3.13)$$

Since the left-hand side of this equation automatically vanishes for $q = \hat{j}$ or $\hat{k}$, the number of conditions is reduced by a factor of 2, and the quaternionic matrix R_{ij} can contain only real and $\hat{i}$ parts and is thus essentially complex. Similarly, the condition (3.4b) on the $(4k+2n)$ -dimensional real vector N becomes

$$\sum_{i=1}^{2k+n} (N_{i,a} + N_{i+2k+n,a} \hat{i}) M_{ib}^C = 0. \quad (3.14)$$

Given a real solution $N_{i,a}$ of Eq. (3.14) for $1 \leq a \leq n$, we can obtain a second solution, defined as $N_{i,a+n}$, by multiplying Eq. (3.14) on the left by $\hat{i}$ so that

$$\left. \begin{aligned} N_{l, a+n} &= -N_{l+2k+n, a} \\ N_{l+2k+n, a+n} &= N_{l, a} \end{aligned} \right\} 1 \leq l \leq 2k+n. \quad (3.15)$$

If the original $N_{l, a}$, $1 \leq a \leq n$ are chosen so that the n complex vectors $(N_{l, a} + N_{l+2k+n, a} \hat{i})$, $1 \leq a \leq n$, are independent with respect to complex multiplication then the resulting $2n$ real vectors $N_{l, a}$, $1 \leq a \leq 2n$, will be linearly independent with respect to real multiplication. With this choice, the real $(4k+2n) \times 2n$ matrix N_{la} can be represented by a $(2k+n) \times n$ complex matrix

$$N_{l, a}^C = N_{l, a} - N_{l+2k+n, a} \hat{i}, \quad \begin{cases} 1 \leq l \leq 2k+n, \\ 1 \leq a \leq n, \end{cases} \quad (3.16)$$

where the last n columns $N_{l, a}$, $n < a \leq 2n$, are not independent of the first n columns when complex multiplication is allowed. The conditions (3.4) become

$$\begin{aligned} N^{C\dagger} N^C &= I, \\ N^{C\dagger} M^C &= 0, \end{aligned} \quad (3.17)$$

while

$$A_\mu^C = N^{C\dagger} \partial_\mu N^C \quad (3.18a)$$

gives an $n \times n$ traceless⁹ anti-Hermitian matrix whose real and imaginary parts make up the original $2n \times 2n$ matrix A of Eq. (3.5),

$$A_\mu = \begin{pmatrix} \text{Re} A_\mu^C & \text{Im} A_\mu^C \\ -\text{Im} A_\mu^C & \text{Re} A_\mu^C \end{pmatrix}. \quad (3.18b)$$

Finally, we note that the projection operator (3.8) is replaced by

$$I - N^C N^{C\dagger} = \text{Re}(M^{C\dagger} R^{-1} M^{C\dagger}) - \hat{i} \text{Re}(M^{C\dagger} \hat{i} R^{-1} M^{C\dagger}). \quad (3.19)$$

Thus, exploiting the extra symmetry that results from restricting $O(2n)$ to $SU(n)$, we obtain the following *Ansatz* for constructing $SU(n)$ instantons: Begin with a $(2k+n) \times k$ matrix of quaternions $M^C(x)$ depending linearly on x and obeying

$$M^{C\dagger}(q - \hat{i}q\hat{i})M^C(x) = 4 \text{Re}(R^\dagger q), \quad (3.20)$$

where R is invertible. Next find a $(2k+n) \times n$ complex matrix N^C obeying

$$N^{C\dagger} N^C = I, \quad (3.21a)$$

$$N^{C\dagger} M^C = 0. \quad (3.21b)$$

Then the equation

$$A_\mu = N^{C\dagger} \partial_\mu N^C \quad (3.22)$$

defines a traceless,⁹ anti-Hermitian, self-dual gauge field which has topological charge k and, as

we will see, depends on the proper number of parameters.

$Sp(n)$. Instanton solutions for the symplectic groups can be constructed from our results for the orthogonal groups by treating $Sp(n)$ as a subgroup of $O(4n)$ in a manner very similar to the above discussion of $SU(n)$. First, recall that $Sp(n)$ can be defined as the group of $n \times n$ matrices Q of quaternions obeying the “unitarity” condition.

$$Q^\dagger Q = I. \quad (3.23)$$

An $n \times n$ quaternionic matrix can be viewed as a transformation on an n -dimensional quaternionic vector space which is linear in the sense that

$$Q(vq) = (Qv)q, \quad (3.24)$$

where v is an n -dimensional quaternionic vector and q is a quaternion. However, such a matrix also describes a real transformation on the $4n$ -dimensional real vector space formed from the $4n$ real numbers necessary to specify the n quaternionic components of v . Conversely, a $4n \times 4n$ real matrix L can be written as an $n \times n$ quaternionic matrix if the analog of the linearity conditions (3.24) is obeyed,

$$LJ_i = J_i L, \quad 1 \leq i \leq 3 \quad (3.25)$$

where the three real matrices $-J_i$ obey the algebra (2.4) of the unit quaternions $\hat{i}$, $\hat{j}$, and $\hat{k}$. If the matrix L obeying Eq. (3.25) is orthogonal, then the corresponding quaternionic matrix will by symplectic, obeying Eq. (3.23).

Adapting this point of view to the problem at hand, we begin with a $(4k+4n) \times k$ matrix $M(x)$, linear in x , satisfying the condition (3.2) and therefore generating an $O(4n)$ self-dual solution. We then impose the additional constraints

$$J_1 M = \hat{i} M, \quad J_2 M = \hat{j} M, \quad (3.26)$$

where, if written in $(k+n) \times (k+n)$ blocks, the matrices J_1 and J_2 are given by

$$J_1 = \begin{pmatrix} 0 & +I & 0 & 0 \\ -I & 0 & 0 & 0 \\ 0 & 0 & 0 & +I \\ 0 & 0 & -I & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 0 & +I & 0 \\ 0 & 0 & 0 & -I \\ -I & 0 & 0 & 0 \\ 0 & +I & 0 & 0 \end{pmatrix}; \quad (3.27)$$

here I is the $(k+n) \times (k+n)$ identity matrix. Thus M can be written in terms of a single $(k+n) \times k$ matrix M^Q :

$$M = \begin{pmatrix} M^Q \\ \hat{i} M^Q \\ \hat{j} M^Q \\ \hat{k} M^Q \end{pmatrix}. \quad (3.28)$$

In terms of the matrix M^Q , the reality condition and definition of R , Eq. (3.2), becomes

$$M^Q(x)^\dagger(q - \hat{i}q\hat{i} - \hat{j}q\hat{j} - \hat{k}q\hat{k})M^Q(x) = 4 \operatorname{Re}(Rq). \quad (3.29)$$

If $\operatorname{Re}(q) = 0$, this condition is automatically obeyed and implies that R is real and thus symmetric. For $q = 1$, it reduces to

$$M^{Q\dagger}(x)M^Q(x) = R. \quad (3.30)$$

Similarly, in terms of M^Q , the condition (3.3b) on the $(4k + 4n) \times 4n$ real matrix N becomes

$$\sum_{i=1}^{k+1} (N_{i,a} + N_{i+k+n,a} \hat{i} + N_{i+2k+2n,a} \hat{j} + N_{i+3k+3n,a} \hat{k}) M_{i,b}^Q = 0, \quad 1 \leq a \leq 4n, \quad 1 \leq b \leq k. \quad (3.31)$$

We define a $(k+n) \times 4n$ quaternionic matrix η by

$$\eta_{i,a} = N_{i,a} - \hat{i}N_{i+k+n,a} - \hat{j}N_{i+2k+2n,a} - \hat{k}N_{i+3k+3n,a}. \quad (3.32)$$

The first n columns of the real matrix N are chosen so that the first n columns of η are independent with respect to quaternionic multiplication. The remaining $3n$ columns of N are then selected by requiring that η be made of four blocks of the form

$$\eta = (N^Q, -N^Q \hat{i}, -N^Q \hat{j}, -N^Q \hat{k}). \quad (3.33)$$

With this choice the matrix N obeys

$$J_i N = N J_i' \quad (3.34)$$

and correspond to the $(k+n) \times n$ quaternionic matrix N^Q . Here J_i' is the obvious $4n \times 4n$ version of the J_i of Eq. (3.27). In terms of the matrices N^Q and M^Q , the projection operator (3.7) takes on the simple form

$$I - N^Q N^{Q\dagger} = M^Q R^{-1} M^{Q\dagger}. \quad (3.35)$$

An $\operatorname{Sp}(n)$, self-dual Yang-Mills field can thus be obtained from a $(k+n) \times k$ matrix M^Q of quaternions linear in x and satisfying the requirement

$$M^Q(x)^\dagger M^Q(x) = R, \quad (3.36)$$

a real, invertible $k \times k$ matrix. The gauge field is given as an $n \times n$ anti-Hermitian matrix of quaternions,

$$A_\mu^Q(x) = N^Q(x)^\dagger \partial_\mu N^Q(x), \quad (3.37)$$

where the $(k+n) \times n$ matrix $N^Q(x)$ obeys

$$N^Q(x)^\dagger M^Q(x) = 0, \quad (3.38)$$

$$N^Q(x)^\dagger N^Q(x) = I.$$

For $n = 1$, this is precisely the formulation of $\operatorname{SU}(2)$ given in Sec. II.

In this way it is possible to obtain $\operatorname{Sp}(n)$ as well as $\operatorname{SU}(n)$ self-dual solutions by suitable restriction of the more general $\operatorname{O}(4n)$ and $\operatorname{O}(2n)$ solutions obtained at the beginning of this section. In both cases, the necessary restrictions somewhat simplify the procedure, resulting in the simplified *Ansätze* (3.36)–(3.38) and (3.20)–(3.22) for $\operatorname{Sp}(n)$ and $\operatorname{SU}(n)$, respectively.

IV. CANONICAL FORM AND COUNTING OF PARAMETERS

In the preceding sections we have seen how a self-dual Yang-Mills field can be constructed quite simply from a matrix M of quaternions obeying a particular nonlinear constraint. It is natural to interpret the physical properties of the gauge field directly in terms of the matrix M . This is most easily done if we exploit the transformations that can be performed on M (but which leave A_μ invariant) in order to put M into a simple canonical form. Using this canonical form the number of parameters entering the k -instanton solution is readily determined. The $\operatorname{O}(n)$ case is considered first and treated in such a manner that the results will be applicable to $\operatorname{SU}(n)$ and $\operatorname{Sp}(n)$.

Let us begin by observing that the gauge field A_μ of Eq. (3.5) is not changed if N and M are transformed according to

$$\begin{aligned} M_{ii} &\rightarrow S_{ij} T_{ji} M_{jk} = M'_{ii}, \\ N_{ii} &\rightarrow S_{ij} N_{ji} = N'_{ii}, \end{aligned} \quad (4.1)$$

where S is a $(4k+n) \times (4k+n)$ real orthogonal matrix and T_{ji} is any invertible $k \times k$ matrix of quaternions. Furthermore, the reality constraint (3.2) on M and the orthogonality of N and M , Eq. (3.4), are unaffected by this transformation. Since the matrices M and M' are completely equivalent we can use the symmetry operations S and T to simplify M . First we choose an orthogonal matrix S which transforms to zero the first n rows of C :

$$S_{ii} C_{ij} = 0, \quad 1 \leq j \leq k, \quad 1 \leq i \leq n. \quad (4.2)$$

Viewing the columns of C as $4k$ real $(4k+n)$ -component vectors, we see that such a matrix must exist.

Next, we write the reality condition (3.2) in terms of the matrices B and C :

$$B^\dagger q B = \text{real}, \quad (4.3a)$$

$$C^\dagger q C = \text{real}, \quad (4.3b)$$

$$B^\dagger q C = (B^\dagger q^\dagger C)^\dagger, \quad (4.3c)$$

for any quaternion q . If we define a $k \times k$ quaternionic matrix $\mathcal{R}$ by

$$C^\dagger q C = 4 \operatorname{Re} q \mathcal{R}^\dagger, \quad (4.4)$$

$$C = \begin{pmatrix} 0 & & & & 0 \\ & \cdot & & & \\ & \cdot & & & \\ & \cdot & & & \\ & \cdot & & & \\ 0 & & \cdot & \cdot & \cdot & 0 \\ 1 & 0 & & & & 0 \\ \hat{i} & 0 & & & & 0 \\ \hat{j} & 0 & \cdot & \cdot & \cdot & 0 \\ \hat{k} & 0 & & & & 0 \\ 0 & 1 & & & & \\ 0 & \hat{i} & & & & \\ 0 & \hat{j} & & & & \\ 0 & \hat{k} & \cdot & & & \\ \cdot & & & \cdot & & \\ \cdot & & & & \cdot & \\ \cdot & & & & & \cdot \\ 0 & & & & & \cdot & 1 \\ 0 & & \cdot & \cdot & \cdot & & \hat{i} \\ 0 & & & & & & \hat{j} \\ 0 & & & & & & \hat{k} \end{pmatrix}$$

FIG. 1. The canonical form for the $(4k+n) \times k$ matrix C .

then $\mathcal{R} = \mathcal{R}^\dagger$ and we can find a $k \times k$ quaternionic matrix T which diagonalizes $\mathcal{R}$,¹⁰

$$T^\dagger \mathcal{R} T = I. \quad (4.5)$$

Thus letting $C'_{ii} = T^\dagger_i C_{ii} T_i$, Eq. (4.4) becomes

$$C'^\dagger q C' = 4I \text{ Reg.} \quad (4.6)$$

Or if we write $C'_{ij} = C'^\mu_{ij} \sigma_\mu$, Eq. (4.6) implies

$$\sum_{l=n+1}^{4k+n} C'^\mu_{il} C'^\nu_{li} = \delta_{ji} \delta_{\mu\nu}, \quad (4.7)$$

so that C'^μ_{ij} itself can be viewed as a $4k \times 4k$ real orthogonal matrix. Our final operation is then an orthogonal transformation on the left of M by

$$\left. \begin{aligned} S_{ii} &= \delta_{ii}, \quad 1 \leq i \leq n \\ S_{ii} &= 0 \\ S_{n+4i-3+\mu, i} &= C'^\mu_{ii} \end{aligned} \right\} 1 \leq i \leq n, \quad n \leq i \leq 4k+n. \quad (4.8)$$

After this transformation, C has the diagonal form

$$C^\mu_{ij} = 0, \quad 1 \leq i \leq n$$

$$C^\mu_{n+4i-3+\mu, j} = \delta_{ij} \delta_{\mu\nu} \quad (4.9)$$

(see Fig. 1). It should be noted that except for a reordering of the rows, this form is consistent with the $O(n)$ description of $SU(n)$ and $Sp(n)$ solutions, having three quarters of the rows of C simple $\hat{i}$, $\hat{j}$, or $\hat{k}$ multiples of the first quarter.

The canonical form (4.9) for C can now be substituted into the reality condition (4.3c), which yields

$$\sum_{\nu=0}^3 [(B^*_{n+4i-3+\nu, j}) \sigma_\mu \sigma_\nu - (B^*_{n+4j-3+\nu, i}) \sigma_\mu^\dagger \sigma_\nu] = 0, \quad (4.10)$$

for $0 \leq \mu \leq 3$. These conditions require

$$\left. \begin{aligned} B_{ab} &= v_{ab}, \quad 1 \leq a \leq n \\ B_{n+4a-3, b} &= b^{(0)}_{ab} + b^{(1)}_{ab} + b^{(2)}_{ab} + b^{(3)}_{ab} \\ B_{n+4a-2, b} &= \hat{i}(b^{(0)}_{ab} - b^{(1)}_{ab}) \\ B_{n+4a-1, b} &= \hat{j}(b^{(0)}_{ab} - b^{(2)}_{ab}) \\ B_{n+4a, b} &= \hat{k}(b^{(0)}_{ab} - b^{(3)}_{ab}) \end{aligned} \right\}, \quad 1 \leq a \leq k \quad (4.11)$$

where $b^{(0)}$ is symmetric and $b^{(1)}$, $b^{(2)}$, and $b^{(3)}$ are antisymmetric $k \times k$ quaternionic matrices, while v is an $n \times k$ quaternionic matrix. Equation (4.10) is obeyed for arbitrary values of the matrices $b^{(r)}$. However, the requirement (4.3a) that $B^\dagger q B$ be real restricts these four matrices so that further non-linear matrix equations must be solved in order to obtain a satisfactory matrix M .

Requiring C to have the canonical form (4.9) does not completely eliminate the freedom to make the transformations (4.1). The choice (4.9) for C is not changed by the transformation (4.1) if the $k \times k$ quaternionic matrix T is symplectic and the $O(4k+n)$ matrix S has the block diagonal form

$$S = \begin{pmatrix} G & 0 \\ 0 & \tilde{T}^{-1} \end{pmatrix}, \quad (4.12)$$

where G is an element of $O(n)$ and $\tilde{T}^{-1}$ is that $4k \times 4k$ orthogonal matrix corresponding to the $n \times n$ symplectic matrix of quaternions T^{-1} .

Having put M into canonical form, we next determine the number of parameters in the corresponding multi-instanton solution. The matrices v and $b^{(r)}$ have altogether $2k^2 - k + nk$ quaternionic elements. However, Eq. (4.3a) requires the reality of one Hermitian (when $q=1$) and three anti-Hermitian matrices (for $q=\hat{i}, \hat{j}, \hat{k}$); this gives $3(2k^2 + k)$ real conditions, so M depends on $2k^2 - 7k + 4nk$ independent parameters. Of these, $\frac{1}{2}n(n-1) + k(2k+1)$ correspond to the $O(n) \times Sp(k)$ group of transformations which preserve the canonical form of C ; subtracting these gives $4(n-2)k - \frac{1}{2}n(n-1)$

true parameters. There are two special cases for which this argument must be modified. The k columns of v may be viewed as $4k$ real n vectors which are acted on by the $O(n)$ transformations. If $n > 4k$, there will be an $O(n - 4k)$ subgroup which leaves these vectors invariant, and thus has no effect on M , so we have oversubtracted by an amount equal to the dimension of this subgroup. Further, if $k = 1$, one can transform v so that it has only four nonvanishing quaternionic components. Some algebra shows that these can be chosen to be c , $\hat{i}c$, $\hat{j}c$, and $\hat{k}c$, where c is some real number. Substituting for G in Eq. (4.12) the choice

$$G = \begin{pmatrix} T^{-1} & 0 \\ 0 & I \end{pmatrix} \quad (4.13)$$

[here I is the $(n - 4) \times (n - 4)$ unit matrix], then gives an $Sp(1)$ subgroup which leaves M invariant, so we must add 3 to our answer. Thus, the number of parameters for the $O(n)$ multi-instanton solution is

$$\begin{aligned} 4(n - 2)k - \frac{1}{2}n(n - 1), \quad k \geq 4n \\ 8k^2 - 6k, \quad 4n \geq k \geq 1 \\ 5, \quad k = 1. \end{aligned} \quad (4.14)$$

A similar analysis can be applied to the $SU(n)$ and $Sp(n)$ constructions. For the unitary case, the matrices S and T of Eq. (4.1) must not affect the $SU(n)$ condition (3.10). Thus the $(4k + 2n) \times (4k + 2n)$ orthogonal matrix S must commute with J and consequently be equivalent to an element of $U(2k + n)$; likewise T must commute with $\hat{i}$ and hence be a complex $k \times k$ matrix. Of course, in the simplified formulation of Eqs. (3.20)–(3.22) the corresponding matrix S would be explicitly a $(2k + n) \times (2k + n)$ unitary matrix. Finally, if we consider $Sp(n)$, the invariance of Eq. (3.26) under the transformation (4.1) requires that S be the real $(4k + 4n) \times (4k + 4n)$ version of an element of $Sp(n)$ while T must be real.

In terms of the matrices $M^C(x)$ or $M^Q(x)$, the canonical form (4.8) and (4.11) adopted for $O(n)$ becomes for $SU(n)$

$$\begin{aligned} C_{ab}^C &= 0, \quad 1 \leq a \leq n \\ C_{n+2a-1, b}^C &= \delta_{ab} \\ C_{n+2a, b}^C &= \hat{j}\delta_{ab} \end{aligned} \left. \vphantom{\begin{aligned} C_{ab}^C &= 0, \quad 1 \leq a \leq n \\ C_{n+2a-1, b}^C &= \delta_{ab} \\ C_{n+2a, b}^C &= \hat{j}\delta_{ab} \end{aligned}} \right\} 1 \leq a \leq k, \quad (4.15)$$

$$\begin{aligned} B_{ab}^C &= v_{ab}, \quad 1 \leq a \leq n \\ B_{n+2a-1, b}^C &= b_{ab}^{(0)} + b_{ab}^{(2)} \\ B_{n+2a, b}^C &= \hat{j}(b_{ab}^{(0)} - b_{ab}^{(2)}) \end{aligned} \left. \vphantom{\begin{aligned} B_{ab}^C &= v_{ab}, \quad 1 \leq a \leq n \\ B_{n+2a-1, b}^C &= b_{ab}^{(0)} + b_{ab}^{(2)} \\ B_{n+2a, b}^C &= \hat{j}(b_{ab}^{(0)} - b_{ab}^{(2)}) \end{aligned}} \right\} 1 \leq a \leq k,$$

while for $Sp(n)$

$$\begin{aligned} C_{ab}^Q &= 0, \quad 1 \leq a \leq n \\ C_{n+2a, b}^Q &= \delta_{ab}, \quad 1 \leq a \leq k \\ B_{ab}^Q &= v_{ab}, \quad 1 \leq a \leq n \\ B_{n+2a, b}^Q &= b_{ab}^{(0)}, \quad 1 \leq a \leq k \end{aligned} \quad (4.16)$$

where, as before, $b^{(0)}$ is symmetric and $b^{(2)}$ anti-symmetric. The group of transformations which preserves the canonical form of C is $U(n) \times U(k)$ for the unitary case and $Sp(n) \times O(k)$ for the symplectic case.

To count parameters for $SU(n)$, we note that M has $k^2 + nk$ quaternionic elements. Equation (3.20) requires the reality of one Hermitian and one anti-Hermitian matrix, giving $3k^2$ real conditions. Subtracting the dimension of $U(n) \times U(k)$ would give $4nk - n^2$ parameters. However, M is invariant under the $U(1)$ subgroup corresponding to $S = e^{-i\Phi}$, $T = e^{-i\Phi}$. Further, the columns of v give $2k$ complex vectors, so if $2k < n$ there is a $U(n - 2k)$ subgroup which has no effect on M . Thus, the number of parameters is

$$\begin{aligned} 4nk - n^2 + 1, \quad 2k \geq n \\ 4k^2 + 1, \quad 2k < n. \end{aligned} \quad (4.17)$$

For $Sp(n)$, M has $kn + \frac{1}{2}k^2 + \frac{1}{2}k$ quaternionic components obeying $\frac{3}{2}k(k - 1)$ real conditions. Subtracting the dimension of $Sp(n) \times O(k)$, and adding the dimension of $Sp(n - k)$ if $k < n$, we obtain

$$\begin{aligned} 4(n + 1)k - n(2n + 1), \quad k \geq n \\ 2k^2 + 3k, \quad k < n. \end{aligned} \quad (4.18)$$

V. $SU(2)$ SOLUTIONS

We now address the problem of explicitly finding matrices $M(x)$ satisfying the nonlinear constraints (4.3). In this section we restrict ourselves to $SU(2) = Sp(1)$; the more general case will be considered in the next section. Let us begin with M in canonical form so that its first row is given by k constant quaternions

$$v_{1j} = M_{1j} = q_j, \quad (5.1)$$

while the remaining k rows of M form a square matrix $\hat{M}(x)$:

$$\hat{M}(x)_{ij} = M_{i+1, j}(x) = \delta_{ij}(y_i - x) + b_{ij}. \quad (5.2)$$

Here the quaternions y_i are the diagonal elements of $\hat{M}$, so that $b_{ii} = 0$. The constraints (4.3) are obeyed if b_{ij} is symmetric and

$$\begin{aligned} \frac{1}{2}(q_i^* q_j - q_j^* q_i) + (y_i - y_j)^* b_{ij} \\ + \frac{1}{2} \sum_{l=1}^k (b_{il}^* b_{lj} - b_{lj}^* b_{il}) = r_{ij} \end{aligned} \quad (5.3)$$

for some real $k \times k$ matrix r .

Finding the general solution to this set of quadratic equations is quite difficult. However, one solution can be obtained immediately. If the q_i are chosen to be real numbers λ_i , then Eq. (5.3) is satisfied if all the b_{ij} vanish. Before proceeding further, let us calculate the gauge field $A_\mu(x)$ corresponding to this simple choice of $M(x)$. We first must find a $(k+1)$ -component quaternionic vector $N(x)$ satisfying

$$N^\dagger(x)M(x) = 0, \quad (5.4a)$$

$$N^\dagger(x)N(x) = 1. \quad (5.4b)$$

If we use Eq. (5.4a) to obtain the last k components of $N(x)$ in terms of the first, we find that the most general solution may be written as

$$N_i = \begin{cases} \frac{1}{\sqrt{\rho}} u, & i=1 \\ -\frac{1}{\sqrt{\rho}} \left[(\hat{M}^\dagger)^{-1} v^\dagger \right]_{i-1} u, & 2 \leq i \leq k+1, \end{cases} \quad (5.5a)$$

where

$$\rho = [1 + v \hat{M}^{-1} (\hat{M}^\dagger)^{-1} v^\dagger] \quad (5.5b)$$

and u is an arbitrary, possibly x -dependent, unit quaternion; different choices for u will give gauge-equivalent Yang-Mills fields. For the simple case we are considering it is trivial to invert $\hat{M}$; we obtain for N

$$N_i(x) = \begin{cases} \frac{1}{\sqrt{\rho_0}} u, & i=1 \\ -\frac{1}{\sqrt{\rho_0}} \frac{y_{i-1} - x}{|y_{i-1} - x|^2} q_{i-1}^* u, & 2 \leq i \leq k+1, \end{cases} \quad (5.6a)$$

where

$$\rho_0 = 1 + \sum_{i=1}^k \frac{\lambda_i^2}{|y_i - x|^2}. \quad (5.6b)$$

If we now make a particularly simple choice for u , namely $u = 1$, we find

$$\begin{aligned} A_\mu(x) &= N^\dagger(x) \partial_\mu N(x) \\ &= -\frac{i}{\rho_0} \bar{\eta}_{a\mu\nu} \sum_{i=1}^k \frac{q_i T^a q_i^* (x - y_i)_\nu}{|x - y_i|^4}, \end{aligned} \quad (5.7)$$

which is the familiar 't Hooft k -instanton solution in “singular gauge.” Here $\bar{\eta}_{a\mu\nu}$ is the parity conjugate of $\eta_{a\mu\nu}$. (Note that this verifies that k is indeed the topological charge.¹¹) The singularities in the gauge field at $x = y_i$ are not physical but are artifacts of our choice $u = 1$.

In order to obtain a more general solution, we

must relax the condition that the q_i be real. Since any quaternion q_i may be written as $(q_i^* q_i)^{1/2} U$ where U is some $SU(2)$ matrix, it is natural to interpret the magnitudes of q_i (which we shall continue to denote by λ_i) as instanton scale parameters and the quaternionic phases of the q_i as specifying the $SU(2)$ orientations of the instantons. An approximate solution to Eq. (5.3) may be obtained for general quaternions q_i if we let the scale parameters λ_i be of order β compared to the “instanton separations” $|y_i - y_j|$ and study the limit of small β . In such a limit the off-diagonal elements b_{ij} become very small and the quadratic piece of the equation can be treated perturbatively. This limit can be described more formally by treating the diagonal elements y_i as fixed and introducing a small parameter β into the q_i i.e., $q_i = \beta q_i^*$, for fixed q_i^* . We must then find the matrix $M(x)$ and the gauge potential $A_\mu(x)$ as a power series in β , choosing the particular branch to the solution of the nonlinear equation (5.3) with the property

$$b_{ij} \sim \beta^2 \quad (5.8)$$

as β tends to zero.

As a first step in this calculation, we exploit the freedom to conjugate $\hat{M}$ by a $k \times k$ orthogonal matrix T to simplify our nonlinear constraint (5.3). For example, if we use

$$T_{ij} = \delta_{ij} + t_{ij} + O(\beta^4), \quad (5.9)$$

where t_{ij} is of order β^2 , to transform M , then to order β^2 , Eq. (5.3) becomes

$$\begin{aligned} \frac{1}{2} (q_i^* q_j - q_j^* q_i) + (y_i - y_j)^* [b_{ij} + (y_i - y_j) t_{ij}] \\ + \frac{1}{2} \sum_{i=1}^k (b_{i1}^* b_{i1} - b_{i1}^* b_{i1}) = r_{ij}. \end{aligned} \quad (5.10)$$

If we choose

$$t_{ij} = \frac{r_{ij}}{|y_i - y_j|^2}, \quad (5.11)$$

then, to order β^2 , r_{ij} drops out of the equation. This procedure can be repeated order by order in β^2 to define an orthogonal matrix T as a power series in β (convergent for β sufficiently small) so that

$$r_{ij} = 0 \quad (5.12)$$

if the constraint (5.3) is written in terms of the elements of the transformed matrix $TM(x)T^{-1}$. With r_{ij} vanishing, Eq. (5.3) can be rewritten

$$b_{ij} = \frac{1}{2} \frac{y_i - y_j}{|y_i - y_j|^2} \left[q_j^* q_i - q_i^* q_j + \sum_i (b_{ij}^* b_{i1} - b_{i1}^* b_{ij}) \right], \quad (5.13)$$

which immediately yields b_{ij} as a power series in β^2 by simple iteration in the quadratic term:

$$b_{ij} = \frac{1}{2} \frac{y_i - y_j}{|y_i - y_j|^2} (q_j^* q_i - q_i^* q_j) + \dots \quad (5.14)$$

The resulting series is convergent for sufficiently small β and gives a solution $M(x)$ containing the full $8k-3$ parameters [$4k$ positions y_i , and $4k-3$ scales and relative SU(2) orientations q_i] which corresponds to k widely separated instantons.

Given this series expansion for $M(x)$, we can again use Eq. (5.5) to explicitly calculate the corresponding gauge field $A_\mu(x)$ for the case of widely separated instantons. It is convenient to make the singular gauge choice $u = 1$. As can be seen from an examination of Eq. (5.5), $N_{j+1}(x)$ can be computed to leading order in β by using an approximate form for $\tilde{M}(x)$ containing only diagonal terms so that the resulting gauge field is simply given by Eq. (5.7). Of course, the q_i in expression (5.7) are no longer real numbers and the terms in Eq. (5.7) nonleading in β are to be ignored.

It is not difficult to find the next correction to Eq. (5.7). In order to describe our approximation precisely, we will discuss the solution $A_\mu(x)$ for two nested regions in position space:

$$\begin{aligned} |x - y_i| >> \lambda_i, & \text{ region I} \\ |x - y_i| >> \beta \lambda_i, & \text{ region II.} \end{aligned} \quad (5.15)$$

Region I consists of points far outside the instantons where the lowest-order solution (5.7) is intrinsically quite small,

$$|A_\mu(x)| \sim \beta^2. \quad (5.16)$$

If we substitute the order- β^2 value for b_{ij} , Eq. (5.14), into Eq. (5.5) for N_{j+1} and use the resulting corrections to N_{j+1} when computing $A_\mu(x)$, we obtain an explicit formula for the gauge potential in region I, including correctly the terms of order β^4 .

$$\begin{aligned} A_\mu(x) = \frac{1}{\rho_0} \sum_{i=1}^k \left\{ -i \tilde{\eta}_{a\mu\nu} Q_i(x) \tau^a Q_i(x)^* \frac{(x - y_i)_\nu}{|x - y_i|^4} \right. \\ \left. + \frac{1}{2} \frac{1}{|x - y_i|^2} [Q_i(x) \partial_\mu Q_i^*(x) - \partial_\mu Q_i(x) Q_i^*(x)] \right\} \\ + O(\beta^6), \end{aligned} \quad (5.17)$$

where

$$Q_i(x) = q_i + \frac{1}{2} \sum_{j \neq i} q_j \frac{(x - y_j)^*}{|x - y_j|^2} \frac{(y_i - y_j)}{|x - y_j|^2} (q_j^* q_i - q_i^* q_j). \quad (5.18)$$

Region II includes all of position space, both inside and outside the instantons, except for very small regions around the singular points y_i . Inside these excluded regions the singular gauge field is so large that corrections of higher order than those being considered can have a significant effect. [Alternatively, our results, Eq. (5.17) and Eq. (5.19), differ from the exact solution by terms vanishing as β^6 and β^2 , *uniformly* in regions I and II, respectively.] Since region II contains points where $|x - y_i| \sim \lambda_i$, the first approximation to the gauge potential, Eq. (5.7), can be of order β^{-1} . The order- β^2 corrections to $\tilde{M}(x)$ can then be used to compute $A_\mu(x)$ accurate through order β everywhere in region II:

$$A_\mu(x) \simeq - \frac{i \tilde{\eta}_{a\mu\nu}}{\rho(x)} \sum_{i=1}^k Q_i(y_i) \tau^a Q_i(y_i)^* \frac{(x - y_i)_\nu}{|x - y_i|^4} + O(\beta^2), \quad (5.19)$$

where

$$\rho(x) = 1 + \sum_{j=1}^k \frac{Q_j(y_j) Q_j(y_j)^*}{|x - y_j|^2}. \quad (5.20)$$

Thus through order β the gauge potential has precisely the form of a superposition of k single instanton solutions. The scale and group orientation of the i th instanton can be obtained by examining Eq. (5.19) in the region $|x - y_i| \sim |q_i|$. The result is simply the scale and SU(2) orientation of a 3rd quaternion $\tilde{Q}_i$,

$$\tilde{Q}_i = q_i - \frac{1}{2} \sum_{j \neq i} \frac{q_j q_i^* q_j}{|y_i - y_j|^2}, \quad (5.21)$$

which differs from q_i by a term of order β^3 . If we express the q_i in terms of a gauge rotation relative to q_i ,

$$q_i = \frac{\lambda_i}{\lambda_i} q_i e^{i \vec{\tau} \cdot \hat{n}_i \theta_i / 2}, \quad (5.22)$$

then

$$\tilde{Q}_i = q_i \left[1 - \frac{1}{2} \sum_j \frac{\lambda_j^2}{|y_i - y_j|^2} (\cos \theta_j + i \hat{n}_j \cdot \vec{\tau} \sin \theta_j) \right]. \quad (5.23)$$

Thus, if there are many randomly oriented instantons, their effects on instanton i will tend to cancel, at least to order β . By contrast, expansion of the 't Hooft solution ($\theta_i = 0$) would have suggested that as the number of instantons became very large the effective scale of each would tend toward zero.

VI. $O(n)$, $SU(n)$, AND $Sp(n)$ SOLUTIONS

We now extend the methods of the previous section to other groups. We begin with the $Sp(n)$ case, since it most closely resembles the $SU(2)$ construction of Sec. V. M is a $(k+n) \times k$ quaternionic matrix, which we assume to be in canonical form. The first n rows form an $(n \times k)$ matrix v_{ij} , while the last k rows form a square matrix $\hat{M}$, which we continue to write in the form (5.2). The constraint (5.3) is replaced by the condition

$$\frac{1}{2} \sum_{i=1}^n (v_{it}^* v_{ij} - v_{ij}^* v_{it}) + (y_i - y_j)^* b_{ij} + \frac{1}{2} \sum_{i=1}^k (b_{it}^* b_{ij} - b_{ij}^* b_{it}) = r_{ij} \quad (6.1)$$

for some real $k \times k$ matrix r .

As with $SU(2)$, one solution can be obtained immediately. If we view the columns of v as n -component vectors, and choose them to be real multiples of a fixed vector e

$$v_{ij} = \lambda_j e_i, \quad (6.2a)$$

then Eq. (6.1) is satisfied if all the b_{ij} vanish. Using the freedom to perform $Sp(n)$ transformations, we can choose

$$e_i = \delta_{i1}. \quad (6.2b)$$

The gauge field $A_\mu(x)$ is then found by a straightforward calculation; it is simply an embedding into $Sp(n)$ of the 't Hooft $SU(2)$ k -instanton solution, with instanton positions y_i and scale parameters λ_i . Furthermore, the embedding is precisely that one which gives the minimum topological charge,⁶ thus verifying that for $Sp(n)$ k is equal to the topological charge.

To obtain more general solutions, we relax the condition that the columns of v be multiples of a fixed vector. Since any n -component quaternionic vector of unit length can be obtained by multiplying the vector e by a suitable $Sp(n)$ matrix G , we can write

$$v_{ij} = \lambda_j (G^{(j)} e)_i, \quad (6.3)$$

where the $G^{(j)}$ are k distinct $Sp(n)$ matrices. We thus obtain a solution with the correct number of physical parameters, namely the y_j , the λ_j , and the parameters needed to describe the relative orientation of the k vectors $G^{(j)} e$. Just as with $SU(2)$, Eq. (6.1) can be analyzed in the limit of widely separated instantons, and these parameters are seen to correspond to the positions, scales and $Sp(n)$ orientations of the instantons.

Let us now proceed to the $O(n)$ and $SU(n)$ cases. We begin by attempting to identify the matrix $M(x)$

corresponding to the embedding of the 't Hooft solution. The results for $Sp(n)$ suggest that we should require [in the notation of Eq. (4.11)]

$$\begin{aligned} b_{ij}^{(0)} &= y_i \delta_{ij}, \\ b_{ij}^{(1)} &= b_{ij}^{(2)} = b_{ij}^{(3)} = 0, \\ v_{ij} &= \lambda_j \tilde{e}_i, \end{aligned} \quad (6.4)$$

where $\tilde{e}$ is fixed vector. Recall, however, that the $O(n)$ and $SU(n)$ constructions impose more conditions on $M(x)$ than does the $Sp(n)$ construction. If we require (6.4), then the $O(n)$ condition (4.3) will be satisfied if

$$\tilde{e}^* q \tilde{e} = \text{real} \quad (6.5a)$$

for any quaternion q , while for $SU(n)$ we need require only

$$\tilde{e}^* \hat{i} \tilde{e} = \text{real}. \quad (6.5b)$$

Making use of those transformations which preserve the canonical form of M , we can write the solutions to these equations as

$$\tilde{e}_i = \frac{1}{2} (\delta_{i1} + \hat{i} \delta_{i2} + \hat{j} \delta_{i3} + \hat{k} \delta_{i4}), \quad (6.6a)$$

for $O(n)$ ($n \geq 4$) and

$$\tilde{e}_i = \frac{1}{\sqrt{2}} (\delta_{i1} + \hat{j} \delta_{i2}) \quad (6.6b)$$

for $SU(n)$. If we now determine the corresponding gauge field, we find that it is indeed an embedding of the 't Hooft $SU(2)$ k -instanton solution with instanton positions y_i and scale parameters λ_i . Just as with $Sp(n)$, the embedding is that which gives the minimum topological charge, thus verifying also for these groups that k is the topological charge.

To go beyond the 't Hooft solution we relax the conditions that the off-diagonal elements of $b^{(r)}$ vanish and that the columns of v be multiples of a fixed vector. In particular, we can consider the case of widely separated instantons and try to solve the analogs of Eq. (6.1) as a power series in β by requiring that the off-diagonal elements of B vanish as β^2 in this limit. Here, as in Sec. V, β parametrizes the ratio of the instanton scales λ_i to the instanton separations $|y_i - y_j|$. As can be seen by examining the terms of order β^2 in Eq. (4.3a), such a solution can exist only if the columns of v each obey a condition of the form (6.5) to order β^2 . Consequently, we can write v in terms of k real numbers λ_j and k $O(n)$ or $SU(n)$ matrices $G^{(j)}$:

$$v_{ij} = \lambda_j (G^{(j)} \tilde{e})_i + O(\beta^2). \quad (6.7)$$

Just as with $Sp(n)$, the physical parameters are seen to correspond to the positions, scales, and group orientations of the instantons.

Note that the solution (6.6a) is not applicable to

O(3). In fact, once Eq. (6.4) is imposed there is no choice of $\tilde{e}$ such that the O(n) condition (4.3) is satisfied. In particular, there is no acceptable $M(x)$ with $k=1$ for O(3), so k cannot be the topological charge. This result could have been anticipated from Eq. (4.14), which gives $4k-3$ as the number of parameters for O(3). This agrees with the SU(2) result only if the topological charge is $k/2$.

VII. CONCLUSION

We conclude with the following remarks:

1. The construction of Atiyah, Hitchin, Drinfeld, and Manin has a simple geometrical interpretation. For example, in the case of SU(2) the matrix $M(x)$ selects a k -dimensional, position-dependent subspace out of a fixed $(k+1)$ -dimensional vector space (here all the dimensions are quaternionic). The vector $N(x)$ lies in the orthogonal complement of the subspace defined by $M(x)$. Given an isospin- $\frac{1}{2}$ field $\psi(x)$, $N(x)$ can be used to map it into this orthogonal complement:

$$\Psi_i(x) = N_i \tilde{\psi}(x), \quad (7.1)$$

where $\tilde{\psi}(x)$ is a quaternion formed from the two isotopic components ψ_1 and ψ_2 of ψ ,

$$\tilde{\psi} = \begin{pmatrix} \psi_1 & -\psi_2^* \\ \psi_2 & \psi_1^* \end{pmatrix}. \quad (7.2)$$

We can define a gauge-covariant derivative of the field Ψ by first taking the ordinary derivative of Ψ and then projecting orthogonally onto the subspace picked out by N . The resulting field $D_\mu \Psi$ then corresponds to the isospin- $\frac{1}{2}$ field $\partial_\mu \psi + A_\mu \psi$.

2. We have seen that the problem of finding self-dual Yang-Mills fields is reduced to that of finding a matrix M satisfying the nonlinear algebraic constraints (4.3). In Secs. V and VI we have solved this problem for the case of widely separated instantons. However, it is clearly desirable to obtain solutions for the general case. Although this is quite difficult for arbitrary topological charge, the general¹² $k=2$ and $k=3$ solutions for SU(2) (in fact for all the symplectic groups) can be found. For $k=2$ the SU(2) constraint Eq. (5.3) is linear in b_{ij} . For any choice of the q_i and y_i it is possible to find an orthogonal matrix T such that r_{12} vanishes; once this has been done, the unique solution to the constraint equation is

$$b_{12} = \frac{1}{2} \frac{(y_1 - y_2)}{|y_1 - y_2|^2} (q_2^* q_1 - q_1^* q_2). \quad (7.3)$$

The y_i and the magnitudes and relative phases of the q_i then give the $8k-3=13$ physical parameters. With $M(x)$ thus determined, substitution into Eq.

(5.5) gives $N(x)$, and thus $A_\mu(x)$, explicitly. Furthermore, by comparing the SU(2) and Sp(n) constraints, Eqs. (5.3) and (6.1), the general $k=2$ result for Sp(n),

$$b_{12} = \frac{1}{2} \frac{(y_1 - y_2)}{|y_1 - y_2|^2} \sum_{i=1}^n (v_{i2}^* v_{i1} - v_{i1}^* v_{i2}), \quad (7.4)$$

is immediately obtained.

To obtain the general $k=3$ solution for SU(2), we first use the SU(2) $\times$ O(k) transformations which preserve the canonical form to require that q_1 be real and that the real parts of the b_{ij} vanish. We then take the y_i and the remaining components of the b_{ij} as the $8k-3=21$ physical parameters. Solving the constraint equation to obtain the q_i in terms of these parameters then gives

$$\begin{aligned} q_1^0 &= \lambda_1 = \frac{|\vec{W}_2 \times \vec{W}_3|}{[\vec{W}_1 \cdot (\vec{W}_2 \times \vec{W}_3)]^{1/2}} \\ q_2 &= \lambda_1 \frac{(\vec{W}_3 \times \vec{W}_2) \cdot (\vec{W}_3 \times \vec{W}_1)}{|\vec{W}_2 \times \vec{W}_3|^2} + i\vec{\tau} \cdot \frac{1}{\lambda_1} \vec{W}_3, \\ q_3 &= \lambda_1 \frac{(\vec{W}_3 \times \vec{W}_2) \cdot (\vec{W}_2 \times \vec{W}_1)}{|\vec{W}_2 \times \vec{W}_3|^2} - i\vec{\tau} \cdot \frac{1}{\lambda_1} \vec{W}_2, \end{aligned} \quad (7.5a)$$

where the vectors $\vec{W}_k$ are defined by

$$\vec{W}_k = \frac{i}{4} \epsilon_{ijk} \text{tr} \left\{ \vec{\tau} \left[(y_i - y_j)^* b_{ij} + \sum_{l=1}^3 (b_{il}^* b_{lj}) \right] \right\}. \quad (7.5b)$$

(Here we have represented the quaternions q_i as 2×2 matrices.) To extend the result to Sp(n) we use the Sp(n) $\times$ O(k) transformations to require that v_{ij} vanish if $l > j$ and be real if $l = j$, and that the real parts of the b_{ij} vanish. Just as with SU(2), the constraints can be used to obtain v_{ij} in terms of the physical parameters. Once $M(x)$ has been obtained, $A_\mu(x)$ can be found explicitly by substitution in Eqs. (5.5) and (2.10).

3. An important step in studying the quantum-mechanical implications of these solutions is to obtain the Green's functions for particles propagating in the presence of a classical self-dual Yang-Mills field. Brown, Carlitz, Creamer, and Lee¹³ have shown that such propagators for massless spinor and vector fields are determined by the corresponding massless scalar propagator. Furthermore, they obtained explicit expressions for the isospin- $\frac{1}{2}$ and -1 massless scalar propagators in the presence of the $5k$ -parameter SU(2) solution of 't Hooft. These expressions can be extended to the general SU(2) solution. For isospin $\frac{1}{2}$ the propagator takes the simple form¹⁴

$$\Delta^{(1/2)}(x, y) = \frac{N^*(x)N(y)}{4\pi^2(x-y)^2}. \quad (7.4)$$

A proof of the formula, making use of the properties of $M(x)$, is given in the Appendix. Note that it is not necessary to specify the gauge in order to obtain the expression; the gauge choice is relevant only in specifying which solution for $N(x)$ is to be chosen. Using the results of the Appendix and following the method of Brown *et al.*, the isospin-1 propagator can also be obtained. It is given by

$$\begin{aligned} \Delta_{ab}^{(1)}(x, y) = & \frac{1}{4\pi^2(x-y)^2} \text{Re}[\tau^a N^\dagger(x) N(y) \tau^b N^\dagger(y) N(x)] \\ & + \frac{1}{8\pi^2} \sum_{ij, mn=1}^k f_{ij, mn} \text{Re}[C^\dagger N(x) \tau^a N^\dagger(x) C]_{ij} \\ & \times \text{Re}[C^\dagger N(y) \tau^b N^\dagger(y) C]_{mn}, \end{aligned} \quad (7.5)$$

where the constants $f_{ij, mn}$ are given by the matrix equation

$$\begin{aligned} f_{ij, mn} L_{mn, rs} = & \delta_{ir} \delta_{js} - \delta_{jr} \delta_{is}, \\ L_{mn, rs} = & \text{Re } 2(C^\dagger B)_{mr} (B^\dagger C)_{sn} - (C^\dagger C)_{mr} (B^\dagger B)_{sn} \\ & - (B^\dagger B)_{mr} (C^\dagger C)_{sn} - (m \leftrightarrow n). \end{aligned} \quad (7.6)$$

These results are quite encouraging, and suggest that the methods developed in this paper may be quite useful in the study of these nonperturbative effects in Yang-Mills theories.

Added note. After this manuscript was completed we received a report by E. J. Corrigan, D. B. Fairlie, S. Templeton, and P. Goddard containing some of these results.

ACKNOWLEDGMENTS

We wish to thank Claude Bernard, Anthony Duncan, Alfred Mueller, and Amarendra Sinha for helpful discussions. The work of N. H. C. and E. J. W. was supported in part by the U. S. Department of Energy, and the work of N. K. S. was sup-

ported in part by the National Science Foundation under Grant No. MCS 76-08478.

APPENDIX

We wish to show that the function

$$\Delta(x, y) = \frac{N^\dagger(x) N(y)}{4\pi^2(x-y)^2} \quad (A1)$$

is in fact the isospin- $\frac{1}{2}$ scalar propagator, i.e., that it satisfies

$$D_\mu D_\mu \Delta(x, y) = -\delta(x-y), \quad (A2)$$

where

$$D_\mu = \partial_\mu + A_\mu(x) = \partial_\mu + N^\dagger(x) \partial_\mu N(x). \quad (A3)$$

After substitution of the form (A1), Eq. (A2) becomes

$$\begin{aligned} \frac{1}{4\pi^2(x-y)^2} \left[-\frac{4(x-y)_\mu}{(x-y)^2} D_\mu N^\dagger(x) N(y) \right. \\ \left. + D_\mu D_\mu N^\dagger(x) N(y) \right] = 0. \end{aligned} \quad (A4)$$

We first note that

$$\begin{aligned} D_\mu N^\dagger(x) = & \partial_\mu N^\dagger(x) [I - N(x) N^\dagger(x)] \\ = & N^\dagger(x) C \sigma_\mu R^{-1}(x) M^\dagger(x), \end{aligned} \quad (A5)$$

where, as in the calculation of $F_{\mu\nu}$ in Sec. II, we have used Eq. (2.14) and the fact that M is linear in x . We next note that

$$\begin{aligned} M^\dagger(x) N(y) = & [M^\dagger(y) - \sigma_\nu^\dagger(x-y)_\nu C^\dagger] N(y) \\ = & -\sigma_\nu^\dagger(x-y)_\nu C^\dagger N(y). \end{aligned} \quad (A6)$$

Then, using

$$\sigma_\mu \sigma_\nu^\dagger = \delta_{\mu\nu} + i\tau^a \eta_{a\mu\nu}, \quad (A7)$$

we obtain

$$-\frac{(x-y)_\mu}{(x-y)^2} D_\mu N^\dagger(x) N(y) = N^\dagger(x) C R^{-1}(x) C^\dagger N(y). \quad (A8)$$

Next, we obtain

$$\begin{aligned} D_\mu D_\mu N^\dagger(x) = & N^\dagger(x) [\partial_\mu + C \sigma_\mu R^{-1}(x) M^\dagger(x)] [C \sigma_\mu R^{-1}(x) M^\dagger(x)] \\ = & N^\dagger(x) \{-4C R^{-1}(x) C^\dagger + C R^{-1}(x) [4C^\dagger M(x) + 2\sigma_\mu M^\dagger C \sigma_\mu] R^{-1} M^\dagger\}. \end{aligned} \quad (A9)$$

For any quaternion q ,

$$\sigma_\mu q \sigma_\mu = -2q^*, \quad (A10)$$

the quaternionic conjugate of q . Using this result

and noting that the constraints (4.3) require that $M^\dagger C$ be symmetric, we see that the last two terms in Eq. (A9) cancel. Substitution of Eqs. (A8) and (A9) into Eq. (A4) then verifies that $\Delta(x, y)$ is indeed the desired propagator.

- ¹V. G. Drinfeld and Yu. I. Manin (unpublished); V. G. Drinfeld and Yu. I. Manin (unpublished).
- ²M. F. Atiyah, N. J. Hitchin, V. G. Drinfeld, and Yu. I. Manin, *Phys. Lett.* **65A**, 185 (1978).
- ³W. Barth, *Inventiones Math.* **42**, 63 (1977).
- ⁴R. S. Ward, *Phys. Lett.* **61A**, 81 (1977).
- ⁵M. F. Atiyah and R. S. Ward, *Commun. Math. Phys.* **55**, 117 (1977).
- ⁶C. W. Bernard, N. H. Christ, A. H. Guth, and E. J. Weinberg, *Phys. Rev. D* **16**, 2967 (1977), and references therein.
- ⁷If B is not of rank k the matrix R defined below is not invertible at $x=0$ and the construction fails. If C has rank less than k , one simply obtains a solution with lower topological charge.
- ⁸G. 't Hooft, *Phys. Rev. D* **14**, 3432 (1976).
- ⁹The gauge field A_μ given by Eq. (3.18) or (3.22) is not automatically traceless. However, it is possible to eliminate the trace of A_μ by a gauge transformation (equivalent to a position-dependent change in the overall phase of N^C) provided $\text{tr}(F_{\mu\nu})=0$. Furthermore, $\text{tr}(F_{\mu\nu})=\partial_\mu(\text{tr} A_\nu)-\partial_\nu(\text{tr} A_\mu)$ corresponds to the field strength of a self-dual, finite action Abelian gauge field and must therefore vanish.
- ¹⁰This congruence of the Hermitian, quaternionic matrix $\mathfrak{R}$ with I is the quaternionic version of a standard theorem in linear algebra for the case of real symmetric or Hermitian matrices. See for example, theorem 9.4 in D. Finkbeiner, *Introduction to Matrices and Linear Transformations* (Freeman, San Francisco, 1960).
- ¹¹Having shown that k is the topological charge for the 't Hooft solution, we can argue that k must be the topological charge for all solutions constructed using our canonical form with M of rank k since these can be obtained from the 't Hooft solutions by continuous deformation of B .
- ¹²A different form for the general $k=2$, $SU(2)$ solution has been previously obtained by conformal transformation of the special 't Hooft $k=2$ solution: R. Jackiw, C. Nohl, and C. Rebbi, *Phys. Rev. D* **15**, 1642 (1977).
- ¹³L. S. Brown, R. D. Carlitz, D. B. Creamer, and C. Lee, *Phys. Rev. D* **17**, 1583 (1978).
- ¹⁴We would like to thank Amarendra Sinha for pointing out to us that the expression found in Ref. 13 could be generalized to this form.

$$C = \begin{pmatrix} 0 & & . & . & . & 0 \\ . & & & & & . \\ . & & & & & . \\ . & & & & & . \\ 0 & & . & . & . & 0 \\ 1 & 0 & & & & 0 \\ \hat{i} & 0 & & & & 0 \\ \hat{j} & 0 & . & . & . & 0 \\ \hat{k} & 0 & & & & 0 \\ 0 & 1 & & & & \\ 0 & \hat{i} & & & & \\ 0 & \hat{j} & & & & \\ 0 & \hat{k} & . & & & \\ . & & & & & \\ . & & & . & & \\ . & & & & & \\ 0 & & & & . & 1 \\ 0 & & . & . & . & \hat{i} \\ 0 & & & & & \hat{j} \\ 0 & & & & & \hat{k} \end{pmatrix}$$

FIG. 1. The canonical form for the $(4k+n) \times k$ matrix C .